

SECTION C

11 (**short**) Consider the following C++ function for solving a quadratic equation:

```
bool solve(float a, float b, float c, float &s1, float &s2)
{
    float d = b*b - 4*a*c;
    if (d<0) return false;
    else {
        s1 = (-b + sqrt(d)) / (2.0*a);
        s2 = (-b - sqrt(d)) / (2.0*a);
        return true;
    }
}
```

- (a) Explain what is indicated by the `bool` value that the function returns. [2]
- (b) Explain the purpose of the two `&` signs in the parameter list. How would the program malfunction if they were omitted? [3]
- (c) When `a`, `b` and `c` are 0.011, 64.1 and 0.011 respectively, the function calculates `s1` as -3.47×10^{-4} , whereas the correct solution is -1.72×10^{-4} . Explain the cause of this large error and suggest how it might be corrected. [5]